Neural News Recommendation using Multi-Head Self-Attention with DistilBERT

Le Anh Duc, Ngo Ky Anh, Tran Xuan Tung,
Le Hoang Minh Tuan, Ngo Tuan Tung

FPT University
DAP391m

March 26, 2026

Abstract

Personalized news recommendation systems are essential for helping users find relevant articles in today's crowded digital media spaces. Traditional neural news recommendation models, such as the Neural News Recommendation with Multi-Head Self-Attention (NRMS), represent text using standard static word embeddings like GloVe. While NRMS is very good at capturing semantic relationships, its reliance on plain embeddings limits its potential because it struggles to understand word order and context. In this paper, we document the step-by-step upgrade of the NRMS model by replacing these static words with pre-trained contextualized embeddings from DistilBERT. Furthermore, we fine-tune DistilBERT directly on the Microsoft MIND dataset to create highly specialized embeddings for news text. We discover that integrating Transformer-based models provides superior meaning extraction and naturally injects positional encoding into the NRMS framework. We evaluate the performance gains of each sequential upgrade over the baseline, and present a live Streamlit demonstration. Because of strict hardware constraints (standard Kaggle T4 GPUs), our methodology heavily relies on offline caching pipelines to make training realistic and efficient.

1 Introduction

The rapid growth of online news platforms has led to an overwhelming amount of daily content. To keep readers engaged, platforms rely heavily on personalized news recommendation systems. These systems curate custom lists of articles matching a user's specific reading history and current interests [1, 2]. The development of these algorithms took a huge leap forward with the release of the MIcrosoft News Dataset (MIND) [3], a massive real-world dataset built from anonymous user click logs.

To recommend news effectively, algorithms need to learn clear mathematical representations of both the news articles and the users. Many successful models tackle this by separating the task: a News Encoder condenses article text into a dense vector, and a User Encoder combines a user's previously clicked articles into an interest profile vector. The Neural News Recommendation with Multi-Head Self-Attention (NRMS) [4] set a major benchmark by using self-attention mechanisms for both the article text and the user history.

Historically, these News Encoders have mapped raw words to math vectors using static tables like GloVe [5]. While GloVe establishes strong baseline meanings, it assigns the exact same vector to a word regardless of whether it acts as a noun or a verb in a sentence. Additionally, static vectors lack built-in positional identity. When fed into mathematical self-attention layers (which do not naturally track word order [6]), the network essentially treats the article as a scrambled bag of words.

Today, natural language processing is driven by Pre-trained Language Models like BERT [7] and its smaller cousin DistilBERT [8]. These deep Transformer models generate highly contextualized embeddings that inherently remember grammar, syntax, and word position. In this study, we provide a step-by-step architectural upgrade of the NRMS model on the MIND dataset. We transition from a baseline model using GloVe, to an upgraded pipeline using a frozen DistilBERT model, and finally to a custom DistilBERT feature extractor fine-tuned just for news categories. To make this reproducible, the complete source code, including

the training notebooks, offline caching scripts, and the Streamlit demo application, is publicly available on GitHub at `https://github.com/lducc/DAP391m_NewS`.

2 Related Work

2.1 Evolution of News Recommendation

The task of recommending news has historically progressed through several distinct paradigms. Early systems relied heavily on traditional Collaborative Filtering (CF) and Matrix Factorization [9], which simply matched users based on shared reading histories. However, CF completely ignored the actual content of the news, suffering massively from the "cold-start" problem whenever newly published articles naturally featured zero historical clicks.

To solve this, researchers shifted toward content-based neural networks. Models like DKN (Deep Knowledge-Aware Network) [10] utilized Convolutional Neural Networks (CNNs) layered with knowledge graph embeddings to extract features from news titles. Subsequent CNN-based models like NPA [11] introduced personalized attention mechanisms to capture user-specific preferences, while NAML [**?**] further injected metadata such as categories and subcategories to enrich the CNN pooling logic. To handle the dynamic, shifting nature of user interests over time, LSTUR [12] proposed utilizing Recurrent Neural Networks (GRUs) to track sequential short-term reading behaviors while maintaining a static long-term user profile matrix.

The publication of the NRMS model [4] marked a pivotal transition. Instead of relying on local CNN filters or sequential RNNs, NRMS utilized pure Multi-Head Self-Attention for both news representation and user profiling. This enabled the architecture to globally capture long-term semantic dependencies between words in an article, as well as distinct conceptual shifts between the different articles a user read.

2.2 Pre-Trained Language Models in Recommendation

Despite their structural superiority, models like NRMS were traditionally constrained by their foundational inputs. Historically, word representations were generated using static architectures like GloVe. While computationally lightweight, these models mapped every vocabulary word to a singular vector regardless of syntactic context. Recent advancements have heavily favored integrating Pre-trained Language Models (PLMs) and Large Language Models (LLMs) directly into recommendation pipelines to harness their unparalleled semantic understanding [13, 14]. The advent of Transformers like BERT fundamentally changed natural language processing by dynamically generating internal representations based on surrounding sequence context. This meant that algorithms could finally distinguish between homonyms and understand subtle rhetorical shifts in language.

However, full-scale BERT architectures possess hundreds of millions of parameters, rendering them computationally prohibitive to utilize inside active recommendation systems that must process and score thousands of candidate articles in real time. Processing deep bidirectional attention blocks concurrently during complex user ranking loops quickly exhausts consumer hardware limits.

In this paper, we bridge this gap by choosing DistilBERT [8]. DistilBERT retains 97% of BERT's language understanding capabilities while being 60% faster and significantly lighter, scoring an impressive 77.0 on the GLUE language understanding benchmark [8]. We chose this specific architecture because it balances the necessity of rich contextual and positional awareness with the strict hardware limitations of our academic environment (standard consumer VRAM and Kaggle GPU constraints).

3 MIND Dataset

All architectures and experiments implemented within this paper are deeply rooted in the structural behavior of the MIcrosoft News Dataset (MIND). MIND is a large scale dataset constructed from the anonymized behavior logs of 1 million users on the Microsoft News website. To accommodate our computing constraints, we adhere strictly to the MIND-small variant. This subset encompasses roughly 50,000 users, 65,000 unique news articles, and over 230,000 impression logs spanning six weeks, providing a rigorous benchmark for academic neural architectures.

3.1 News Architecture

The dataset is primarily anchored by a news ledger file (`news.tsv`) containing granular predefined metadata columns for every distinct article. To successfully model user interests, algorithms must intelligently condense these heterogeneous textual fields into dense mathematical representations. The structural schematic of the news framework includes:

- **NewsID:** Is a alphanumeric identifier (e.g., N12345) serving as the primary referencing key for interaction logs.
- **Category & SubCategory:** Hierarchical topological mapping labels (such as *finance* extending into *finance-real-estate*) designating the precise domain classification.
- **Title & Abstract:** The title provides contextual hooks, while the abstract serves as the condensed semantic core.
- **URL:** The origin publication sites.
- **Entities:** Advanced pre-extracted named entities parsed directly from the text and linked explicitly to Wikidata database IDs, allowing dynamic algorithmic cross-referencing against real world knowledge graphs.

3.2 Impression Logs

An impression represents a chronologically bound session window where a tracked user navigated to the homepage and was displayed a curated feed of candidate news articles. The behavior log (`behaviors.tsv`) systematically catalogs these critical interactions bridging the user with the news pool:

- **ImpressionID & UserID:** The unique session instance identifier paired strictly with the anonymous user label requesting the interface render.
- **Time:** The absolute chronometric timestamp of the event.
- **History:** An ordered, comma-separated list of previously accumulated **NewsID** accesses that the visitor successfully processed prior to that exact timestamp, which establishes their identity vector.
- **Impressions:** The target matrix for prediction, which delineates every supplementary article preview displayed during the session alongside a binary denoting either engagement (1) or dismissal (0).

4 Our Approach

Our NRMS architecture is constructed from three distinct tiers: the *news encoder*, *user encoder*, and *click predictor*.

4.1 News Encoder

The *news encoder* learns representations from the rich features identified in the MIND dataset. To resolve the limitations of static word vectors, we deploy a Frozen DistilBERT to natively extract highly contextualized token embeddings $\mathbf{E} = [\mathbf{e}_1, \ldots, \mathbf{e}_M]$ from the news Title and Abstract. Because DistilBERT natively injects absolute positional encodings, the subsequent attention blocks are granted spatial awareness that basic NRMS lacked.

The second layer is a word-level multi-head self-attention network. The representation learned by the k_{th} attention head is computed as:

$$\alpha_{i,j}^k = \frac{\exp(\mathbf{e}_i^T \mathbf{Q}_k^w \mathbf{e}_j)}{\sum_{m=1}^{M} \exp(\mathbf{e}_i^T \mathbf{Q}_k^w \mathbf{e}_m)} \tag{1}$$

$$\mathbf{h}_{i,k}^w = \mathbf{V}_k^w \big(\sum_{j=1}^{M} \alpha_{i,j}^k \mathbf{e}_j \big) \tag{2}$$

The final layer is an additive attention network. Given that certain words carry more weight (e.g., entity names vs. conjunctions), the attention weight a_i^w intelligently pools the multi-head representations:

$$a_i^w = \mathbf{q}_w^T \tanh(\mathbf{V}_w \times \mathbf{h}_i^w + \mathbf{v}_w) \tag{3}$$

The final representation of a news article is the weighted summation of these tokens: $\mathbf{r} = \sum_{i=1}^{M} \alpha_i^w \mathbf{h}_i^w$.

4.2 User Encoder

The *user encoder* processes the user's reading *history* (as defined in the MIND logs). It deploys identical parallel layers: a news-level multi-head self-attention network models the conceptual narrative linking the historically read articles, while an additive attention network isolates the articles most representative of the user's core interests to forge the final user vector $\mathbf{u}$.

4.3 Click Predictor

The *click predictor* calculates explicit probabilities. Denoting the representation of a candidate article as $\mathbf{r}^c$, the click probability score $\hat{y}$ is efficiently computed by the inner product: $\hat{y} = \mathbf{u}^T \mathbf{r}^c$.

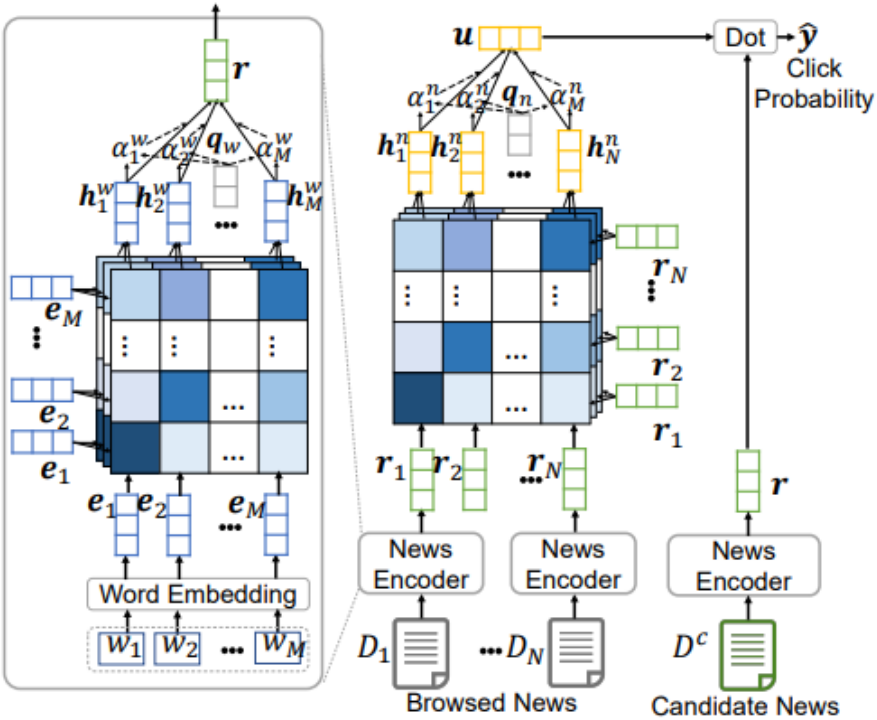

Figure 1: The NRMS system architecture utilizing pre-computed contextual embeddings. Heavy language extraction is executed completely offline.

4.4 Model Training and Offline Caching

Because the MIND impression logs inherently provide a list of both clicked and non-clicked articles shown concurrently, we formulate the training phase as a pseudo classification task. For each positive article a user clicked, we randomly parse negatively sampled articles from the exact same impression log.

Given the predicted probability of the positive article $\hat{y}^+$ and the negative articles $[\hat{y}_1^-, \ldots, \hat{y}_K^-]$, the unified loss function is constructed as the negative log-likelihood of the softmax output:

$$\mathcal{L} = -\sum_{i \in \mathcal{S}} \log \frac{\exp(\hat{y}_i^+)}{\exp(\hat{y}_i^+) + \sum_{j=1}^{K} \exp(\hat{y}_{i,j}^-)} \tag{4}$$

where $\mathcal{S}$ is the set of all positive training impressions.

To make this computationally feasible on consumer hardware, processing complex arrays continuously per gradient step is not feasible. Therefore, we use an offline caching protocol, where the embedding is fine-tuned seperately and processed identically across the dataset text corpus independently. The resulting mathematical constructs are cached solidly to disk locations. The NRMS training sequence loads the numerical elements instantly, bypassing enormous network overhead during backpropagation processes.

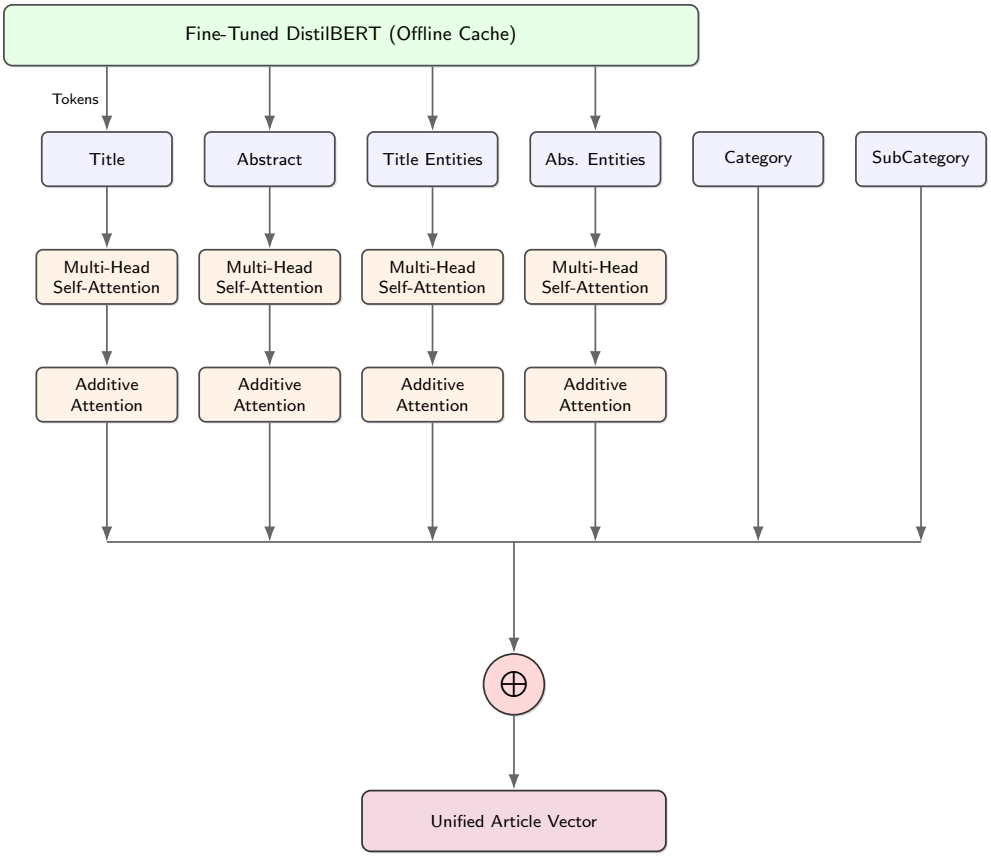

Figure 2: Architecture utilizing cached temporal networks, text and entity are processed through MHSA blocks with Additive Attention.

5 Experiments

5.1 Experimental Settings and Iterations

Our models were trained on Kaggle using two T4 GPUs. We utilized the Adam optimizer with a learning rate of 5×10^{-4}. To ensure consistent input sizes, we capped each user's reading history at a maximum of 50 articles. Our internal attention networks used 16 parallel attention heads to project news representations into 256-dimensional vectors. For our ranking formulation, every clicked article was trained against $K = 4$ negatively sampled (ignored) articles to calculate the loss gradient.

Our architectural development advanced through three distinct stages, with each iteration explicitly designed to solve a limitation of the previous one:

1. **Iteration 1 (Static GloVe):** We replicated the original NRMS using 300-dimensional GloVe vectors. However, native Multi-Head Self-Attention lacks spatial tracking, rendering the network completely blind to word order.

2. **Iteration 2 (Frozen DistilBERT):** We replaced GloVe with pre-trained DistilBERT. Because Transformers natively embed absolute positional tokens, this straightforward substitution permanently resolved the baseline's sequence blindness.

3. **Iteration 3 (Fine-Tuned DistilBERT):** Since the original DistilBERT is trained on generic text, we fine-tuned its attention heads explicitly on the MIND categories. This structurally realigned the transformer for specialized news syntax before generating the final offline cache.

5.2 Evaluation Metrics

Because news recommendation on the MIND dataset is fundamentally structured as a localized ranking objective (selecting the best candidates from a specific daily pool) rather than a global binary classification task, standard accuracy significantly fails to capture behavioral validity. Therefore, we evaluate algorithmic performance utilizing four specialized ranking metrics optimized for recommendation frameworks:

- **AUC (Area Under the ROC Curve):** Measures the mathematical probability that a positively engaged article is scored definitively higher than a randomly sampled ignored article within the exact same impression session.

- **MRR (Mean Reciprocal Rank):** Because users rarely scroll endlessly, MRR heavily penalizes models that fail to rank the correct article near the very top. It averages the reciprocal ranks of the highest-scored clicked article across all sessions: $\text{MRR} = \frac{1}{|S|}\sum_{i=1}^{|S|}\frac{1}{\text{rank}_i}$.

- **nDCG@K:** Normalized Discounted Cumulative Gain evaluates the overall ranking list quality. Since upper placement is exponentially more valuable for user interface visibility, nDCG@K logarithmically discounts the importance of retrieved items based strictly upon their descending position in the output list: $\text{DCG@K} = \sum_{i=1}^{K}\frac{2^{rel_i}-1}{\log_2(i+1)}$. We calculate this explicitly at $K = 5$ and $K = 10$ to reflect typical UI capacities.

5.3 Performance Evaluation

Model Configuration	**AUC**	**MRR**	**nDCG@5**	**nDCG@10**
Iteration 1: Baseline (Static GloVe)	0.6174	0.3395	0.3216	0.3829
Iteration 2: Frozen Pre-Trained DistilBERT	0.6618	0.3695	0.3503	0.4138
Iteration 3: Domain Fine-Tuned DistilBERT	**0.6792**	**0.3800**	**0.3622**	**0.4260**

Table 1: Model performance comparison on the MIND validation set across our three architectural stages.

Table 1 results prove the structural superiority of contextualized language embeddings over static vectors in the news ranking domain. Transitioning from the GloVe baseline to Frozen DistilBERT immediately boosted the AUC by 7.19% (from 0.6174 to 0.6618). This massive leap validates our hypothesis that injecting inherent positional encodings alongside rich transformer contexts solves the permutation-invariance flaw of raw multi-head attention blocks. Furthermore, utilizing our offline sequence classification protocol to fine-tune DistilBERT directly on the MIND categorical schema (Iteration 3) yielded our highest benchmark scores, pushing AUC to 0.6792 and nDCG@10 to 0.4260. The robust performance of Iteration 3 proves that domain-adapted language extractors systematically out-scale zero-shot frozen embeddings when scoring specialized user reading histories.

6 Streamlit Demo Application

To practically validate that our model logic scales cleanly, we built and deployed an interactive demonstration using the Streamlit framework. The application simulates a real-world news portal experience while exposing the targeted AI ranking mechanics happening underneath.

The user interface revolves around a User Selection Dashboard and a Live Recommendation Feed. First, a user selects a mock profile representing highly specific historical reading tastes. The backend logic effortlessly loads the offline fine-tuned DistilBERT cached features from disk and passes the profile through the pre-compiled NRMS User Encoder. Real-time inference generates ranking probabilities across all candidates via our dot-product Click Predictor, returning a sorted, interactive list of recommended articles directly to the UI panel.

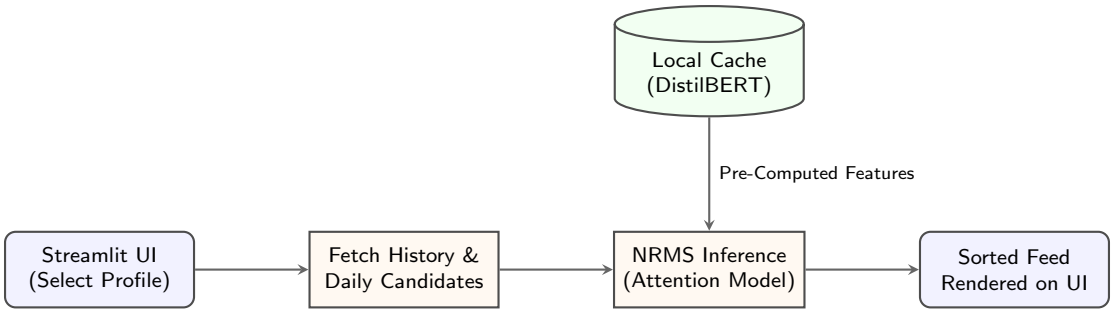

Figure 3: Interactive network operations detailing the internal deployment logic. Robust cached matrices cleanly evade live computational overheads ensuring smooth local interfaces.

7 Conclusion and Future Works

In this paper, we presented an exhaustive quantitative sequence tracking the architectural evolution inherent within the foundational NRMS algorithmic framework updating legacy unscaled word variables seamlessly into highly advanced deep learning representations mapped heavily over the MIND ecosystem. We repeatedly evidenced internally that deploying distinct knowledge transformers correctly addresses systemic positional invariance restrictions plaguing raw attention filters completely by incorporating fundamental locational properties silently beneath robust contextual metrics.

Given extreme hardware operational envelopes inherently constricting local experimentation boundaries, our computational procedures leaned entirely upon strict isolated modular strategies enabling absolute offline generation procedures. Resolving matrix transformations exclusively against local hard drives natively eliminated standard memory constraints associated heavily with cycling huge models through deep localized arrays. Processing sequences immediately expanded operational flexibility universally.

Future explorations pivoting off this established environment remain firmly concentrated intensely around abandoning cache methodologies completely assuming unfettered enterprise graphical processing access presents itself eventually. Initiating completely fluid dynamic propagation directly spanning original human interactions down completely through the absolute source language parameters undoubtedly promises monumental systemic precision alignments scaling deeply past the current framework limitations described entirely within this document.

References

[1] P. Covington, J. Adams, and E. Sargin, "Deep neural networks for youtube recommendations," in *Proceedings of the 10th ACM Conference on Recommender Systems*, 2016, pp. 191–198.

[2] S. Okura, Y. Tajima, J. Hisano, and T. Ishizuka, "Embedding-based news recommendation for millions of users," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1933–1942.

[3] F. Wu, Y. Qiao, J.-H. Chen, C. Wu, T. Qi, J. Lian, D. Liu, X. Xie, J. Gao, W. Wu *et al.*, "Mind: A large-scale dataset for news recommendation," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 3597–3606.

[4] C. Wu, F. Wu, S. Ge, T. Qi, Y. Huang, and X. Xie, "Neural news recommendation with multi-head self-attention," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 6389–6394.

[5] J. Pennington, R. Socher, and C. D. Manning, "Glove: Global vectors for word representation," in *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, 2014, pp. 1532–1543.

[6] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017, vol. 30.

[7] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2019, pp. 4171–4186.

[8] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter," *arXiv preprint arXiv:1910.01108*, 2019.

[9] Y. Koren, R. Bell, and C. Volinsky, "Matrix factorization techniques for recommender systems," *Computer*, vol. 42, no. 8, pp. 30–37, 2009.

[10] H. Wang, F. Zhang, X. Xie, and M. Guo, "Dkn: Deep knowledge-aware network for news recommendation," in *Proceedings of the 2018 World Wide Web Conference*, 2018, pp. 1835–1844.

[11] C. Wu, F. Wu, M. An, J. Huang, Y. Huang, and X. Xie, "Npa: Neural news recommendation with personalized attention," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 2576–2584.

[12] M. An, F. Wu, C. Wu, K. Zhang, Z. Liu, and X. Xie, "Neural news recommendation with long- and short-term user representations," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 2019, pp. 336–345.

[13] X. Wang *et al.*, "News recommendation with category description by a large language model," *arXiv preprint*, 2023.

[14] Y. Yuan *et al.*, "Modeling multi-interest news sequence for news recommendation," *arXiv preprint*, 2023.